

Restaurant Sales and Inflation

A plain-language business analytics project report

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The research question

How much of the growth in U.S. food services and drinking places sales remains after adjusting for changes in the price of food away from home?

What was done

Two monthly public data series were joined by date: Census sales and a BLS consumer price index, both downloaded through FRED. Sales were converted to the January 2019 price level. The project compares reported dollars, adjusted dollars, indexed trends, and growth over the same month a year earlier.

Three findings

- July 2026 reported sales were \$103.82 billion, compared with approximately \$73.35 billion expressed at January 2019 prices. These values use different dollar bases.
- From July 2025 to July 2026, reported sales increased 5.20%, the food-away-from-home price index increased 3.40%, and adjusted sales increased 1.74%.
- Compared with January 2019, reported sales were 72.48% higher; adjusted sales were 21.86% higher. The larger reported increase should not be read as an equivalent increase in activity.

Scope and verification

This is a descriptive quantitative study, not a forecasting or causal model. The calculations were independently replicated from the saved CSV files and checked against the supplied RStudio console results. No interviews, survey narratives, or qualitative coding were conducted.

1. Data and analytical method

Source data

Series	Measure and units	Saved coverage
MRTSSM722USS	Food services and drinking places sales; millions of dollars; seasonally adjusted	Jan 2010–Jul 2026; 199 rows
CUSR0000SEFV	Food away from home CPI; 1982–84 = 100; seasonally adjusted	Jan 2010–Aug 2026; 200 rows

Sales come from the U.S. Census Bureau; prices come from the Bureau of Labor Statistics. “Food services and drinking places” includes restaurants and bars. The price index has related, but not identical, coverage, so it is an approximate deflator for this revenue series.

Saving and cleaning the data

Original downloads were preserved in data/raw. Dates were converted to calendar dates, and monthly keys were checked for duplicates. A left join retained all 199 sales months. The extra August 2026 CPI month was outside the available sales period and was not added to the analysis.

October 2025 CPI is blank in the saved snapshot. Its reported sales remain available, but its adjusted sales and price-dependent growth measures remain missing. No values were filled in. There are 198 usable sales–price pairs.

The inflation adjustment

Adjusted sales = reported sales $\times$ (January 2019 CPI $\div$ current-month CPI). The reference CPI is 280.380. For July 2026: $103,824 \times (280.380 \div 396.859) = 73,351.42$ million January 2019 dollars.

Both indexed series divide their dollar value by January 2019 sales of \$60.194 billion and multiply by 100. The reference month therefore equals 100 for both lines.

Comparing growth

Year-over-year growth = (current month $\div$ the same calendar month one year earlier $- 1$) $\times 100$. Exact date matching prevents missing observations from shifting the comparison to the wrong month. Seasonally adjusted values were used as supplied.

2. Inflation-adjusted sales over time

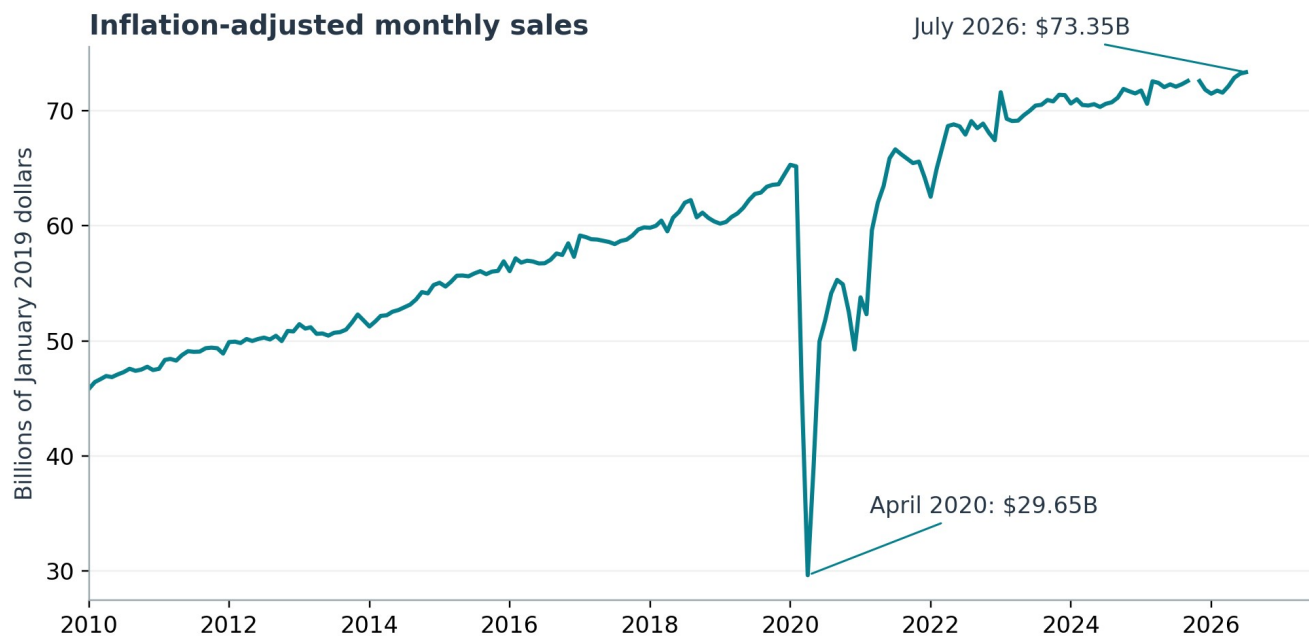


Figure 1. Monthly sales expressed at January 2019 prices. Source: saved Census sales and BLS CPI series through FRED; independent calculation. October 2025 remains missing.

How to read this figure

The horizontal axis shows time. The vertical axis shows billions of dollars at one fixed price level. This makes dollar amounts from different years more comparable. The line is not a count of customers or meals.

What the data show

Adjusted sales fell to about \$29.65 billion in April 2020 and later recovered. By July 2026, they reached approximately \$73.35 billion. January 2019 adjusted sales were \$60.19 billion, making the latest value 21.86% higher than that reference month.

Why the missing month matters

The adjusted series contains a gap in October 2025 because its CPI observation is unavailable. Keeping a missing value is more transparent than substituting zero or silently estimating a price. It also preserves the distinction between observed and estimated data.

The plot describes the movement in the series. It does not, on its own, establish why sales changed or separate changes in customer counts, product mix, or spending per purchase.

3. Reported versus adjusted growth

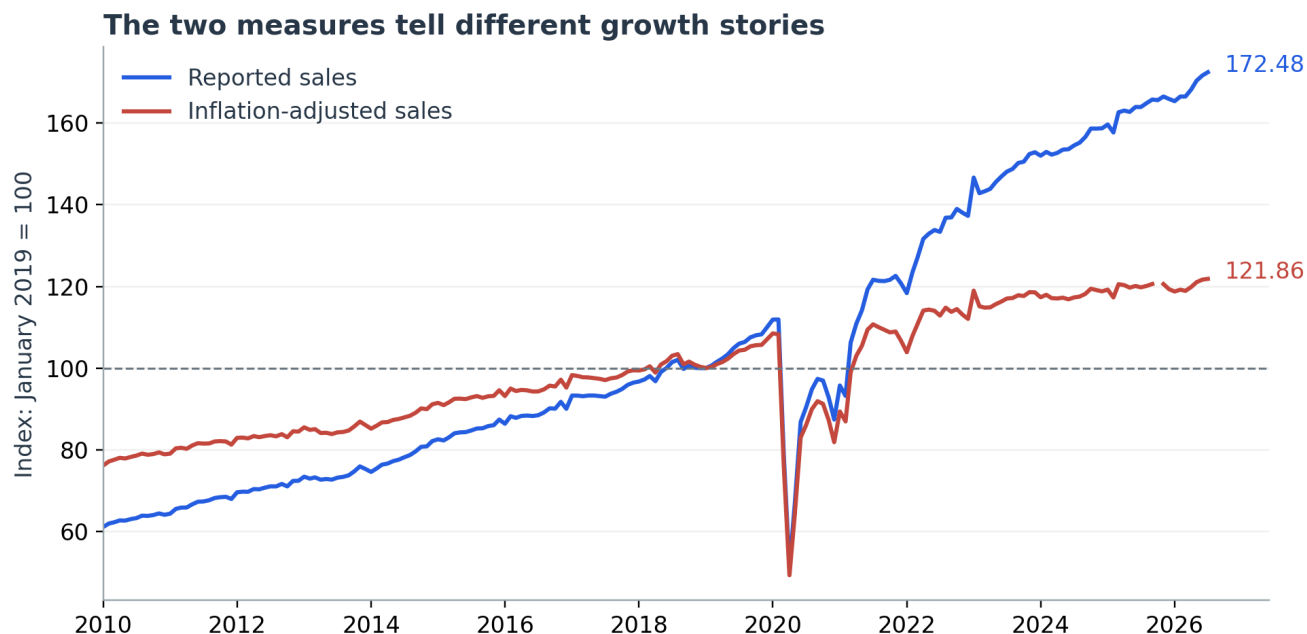


Figure 2. Both series use January 2019 = 100. Blue shows reported sales; red shows sales at the January 2019 price level. Source: saved FRED data; independent calculation.

How to read the index

An index of 120 means the value is 20% above January 2019. The reference month is a comparison point, not the start of the dataset. Before 2019, adjusted dollars can exceed reported dollars because the adjustment expresses them at the later reference price level.

Month	Reported sales	Adjusted sales	Reported / adjusted index
Jan 2019	\$60.19B	\$60.19B	100.00 / 100.00
Jul 2025	\$98.69B	\$72.10B	163.95 / 119.77
Jul 2026	\$103.82B	\$73.35B	172.48 / 121.86

The business interpretation

The July 2026 reported index is 172.48, while the adjusted index is 121.86. The separation shows why revenue growth and growth after a price adjustment are different measures. A business comparing its performance with this industry series should specify which measure it uses.

4. The latest year-over-year comparison

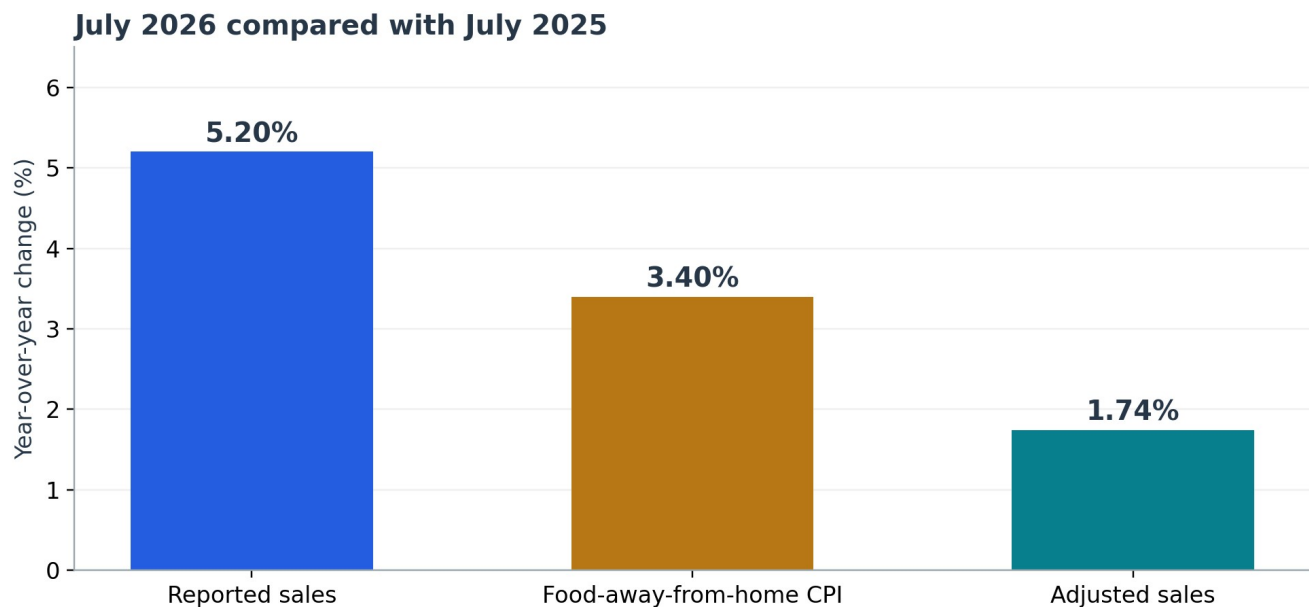


Figure 3. Changes between July 2025 and July 2026. Source: saved FRED data; independent calculation. Bars show separate measures, not an additive decomposition.

What the three numbers mean

Reported revenue grew 5.20%. The relevant price index grew 3.40%. After applying the CPI adjustment, sales grew 1.74%. Growth remained positive, but was smaller when measured at a fixed price level.

Why we use a ratio instead of subtraction

Adjusted growth = $[(1 + \text{reported growth}) \div (1 + \text{price growth}) - 1] \times 100$, using growth rates as decimals inside the formula. For this comparison, $(1.0520108 \div 1.0340040 - 1) \times 100 = 1.7415\%$.

Subtracting 3.40 from 5.20 gives an approximation of 1.80%, rather than the exact adjusted result. The 3.46 percentage-point gap between reported and adjusted growth is not itself the inflation rate or a causal estimate of inflation's share of growth.

Which month is “latest”?

July 2026 is the final sales month in the saved data. Although CPI extends to August 2026, an August adjusted sales result cannot be calculated without a corresponding sales observation.

5. Discussion, limits, and conclusion

What a business can learn

A revenue dashboard can show both reported and adjusted sales so users can distinguish dollar growth from growth after a price adjustment. The same-month comparison provides an industry context for changes in a business's own sales. A company-specific analysis would also need its own prices, transaction counts, product mix, and comparable-store data.

What this analysis cannot establish

- Adjusted revenue is an approximation, not a direct measure of visits, meals sold, or physical output. Prices and the composition of purchases can change together.
- Industry revenue does not measure profits. This project contains no labor, rent, food-input, or other operating-cost data.
- The sales series covers food services and drinking places, while the CPI measures consumer food-away-from-home prices. Their coverage is not identical.
- A national aggregate cannot establish how a particular restaurant, region, or customer group performed. It is not a same-store measure.
- The data are a saved snapshot and may be revised. Results describe these saved values rather than a historical real-time data vintage.
- No causal identification, forecasting model, or statistical significance test was used. The charts and calculations describe observed changes.

Does this include qualitative research?

The data, calculations, and charts provide a quantitative component. Written interpretation explains the numbers but is not a qualitative research component. A mixed-methods assignment would additionally require systematically collected and analyzed text, interviews, or other qualitative evidence. Those materials are outside this completed project.

Conclusion

Restaurant-industry revenue grew faster in reported dollars than after the food-away-from-home price adjustment. In July 2026, year-over-year adjusted growth was 1.74%, compared with 5.20% reported growth. Showing both measures gives a more useful description of sales performance than reporting nominal revenue alone.

6. Reproducing the work and sources

The RStudio workflow

1. Unzip the project and open RStudio. Set the working directory to the extracted restaurant_project_final folder, which contains analyze_restaurant_sales.R.
2. Run source("analyze_restaurant_sales.R"). The script uses base R and requires no additional packages.
3. The script reads the preserved raw CSV files, joins dates, calculates adjusted sales and growth, checks the results, and saves processed CSV files and three charts in outputs.
4. Compare outputs/latest_month_summary.csv with the findings in this report. Retain the raw files so the same snapshot can be reproduced.

Checks performed for this report

Check	Result
Date keys and join	Unique monthly keys; 199 sales rows retained
Missingness	October 2025 CPI and adjusted sales remain missing
Reference and arithmetic	Jan 2019 equals 100; reverse adjustment recovers reported sales
Year-over-year calculation	Exact prior-year dates; growth ratio identity verified
Latest-month comparison	Matches supplied R console values after rounding

Report calculations and charts were independently generated in Python from the saved files. R was unavailable in the report-building environment, so the supplied final R script was reviewed but not executed there. It reproduces the same formulas and provides R chart exports with “_R” filenames. Full validation details and raw-file checksums are saved in outputs/validation_record.json.

Project files

data/raw: original sales and CPI downloads. data/processed: joined analysis table. outputs: charts, latest-month summary, missing-month record, and verification record. docs: this report. analyze_restaurant_sales.R: reproducible R analysis.

Sources and documentation

Federal Reserve Bank of St. Louis, FRED. Advance Retail Sales: Food Services and Drinking Places (MRTSSM722USS). Source agency: U.S. Census Bureau.
<https://fred.stlouisfed.org/series/MRTSSM722USS>

Federal Reserve Bank of St. Louis, FRED. Consumer Price Index for All Urban Consumers: Food Away from Home in U.S. City Average (CUSR0000SEFV). Source agency: U.S. Bureau of Labor Statistics.
<https://fred.stlouisfed.org/series/CUSR0000SEFV>

Series definitions and current page metadata checked October 1, 2026. Computation uses the saved September 29, 2026 snapshot.